

Movie Recommendation System

Improving User Engagement & Retention
Through Personalization

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Stakeholders & Value

- ▶ **Primary Stakeholders**
- ▶ Streaming platform management
- ▶ Product and marketing teams
- ▶ Customer experience teams
- ▶ **Value to Stakeholders**
- ▶ Higher user engagement
- ▶ Improved retention
- ▶ Better use of existing content

Overview

- ▶ Predict movies users are likely to enjoy
- ▶ Provide **Top-5 personalized recommendations**
- ▶ Ensure recommendations work for:
 - ▶ Existing users
 - ▶ New users
 - ▶ Newly added movies
- ▶ Support long-term engagement and retention

Business Understanding

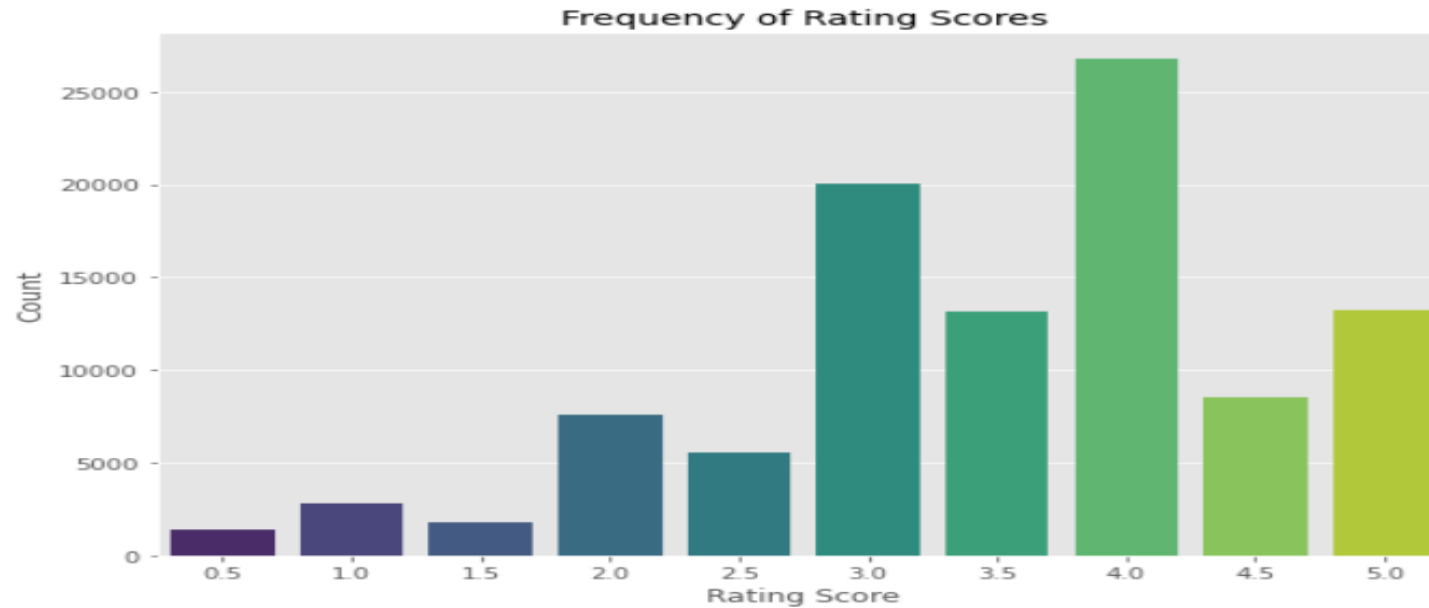
- ▶ **The Challenge**
- ▶ Streaming platforms offer thousands of movies
- ▶ Users struggle to find content they will enjoy
- ▶ Poor recommendations result in:
 - ▶ Reduced watch time
 - ▶ Lower user satisfaction
 - ▶ Higher user churn
- ▶ **Business Need:**
Deliver accurate and personalized movie recommendations.

Data Understanding

- ▶ **Data Used**
- ▶ Real user movie ratings (1-5 stars)
- ▶ Movie information:
 - ▶ Titles
 - ▶ Genres
- ▶ **Dataset Size**
- ▶ 610 users
- ▶ Over 9,700 movies
- ▶ Over 100,000 ratings
- ▶ **Why This Matters**
- ▶ Reflects real user preferences
- ▶ Suitable for personalization

Key Insight from the Data

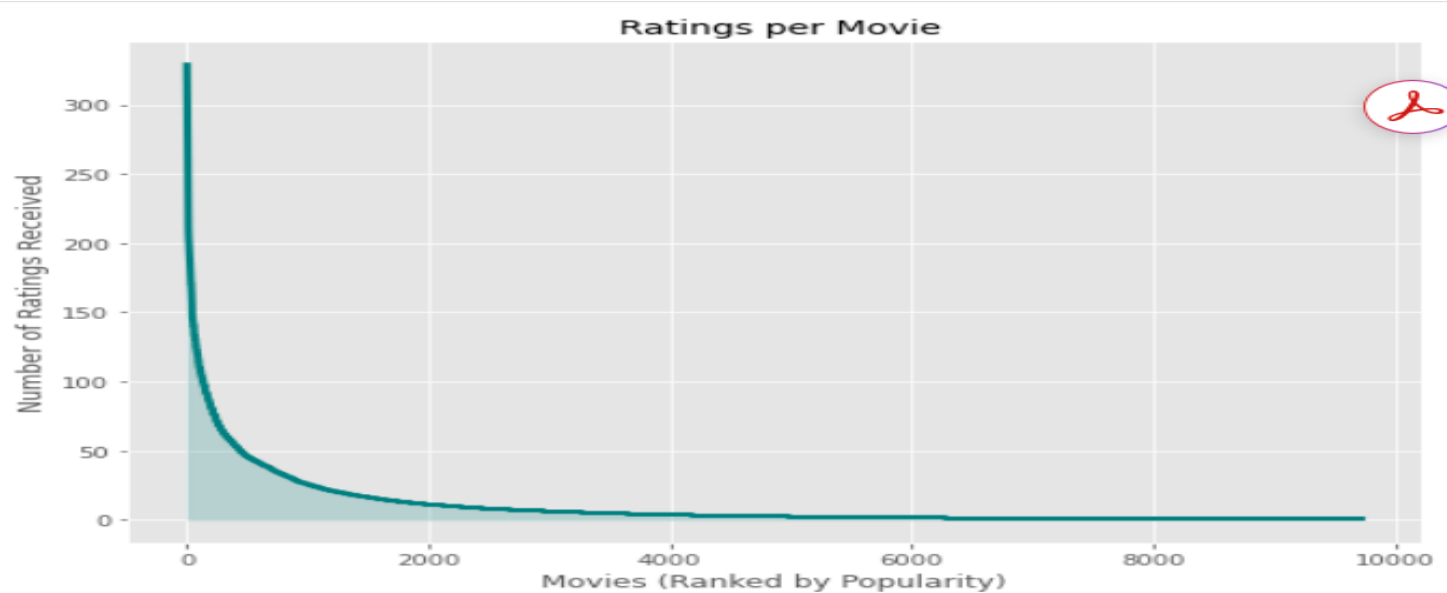
How Users Typically Rate Movies



- ▶ Most ratings fall between 3 and 4 stars
- ▶ Users usually rate movies they already like
- ▶ Indicates generally positive engagement
- ▶ **Caption (below graph):**
User ratings are mostly positive, highlighting the importance of recommending relevant content.

Data Challenge

Not All Movies Are Rated Equally



- ▶ A few popular movies receive most ratings
- ▶ Many movies have very limited feedback
- ▶ This makes recommendation challenging
- ▶ **Caption (below graph):**
User feedback is unevenly distributed, requiring intelligent prediction methods.

Key Insights from the Data

- ▶ Most ratings fall between **3 and 4 stars**
- ▶ Users rate only a small fraction of available movies
- ▶ A few movies are very popular, most are rarely rated

Implication

- ▶ The system must infer preferences from limited data

Recommendation Strategy

- ▶ The system uses a **hybrid recommendation approach**:
- ▶ Learns from user viewing and rating behavior
- ▶ Identifies similarities between movies
- ▶ Combines multiple methods to ensure reliability

Outcome

- ▶ Personalized, relevant movie suggestions

Handling New Users & Movies

Solving the Cold-Start Problem

- ▶ Existing users → personalized recommendations
- ▶ New users → genre-based suggestions
- ▶ New movies → content similarity

Business Benefit

- ▶ Every user receives meaningful recommendations
- ▶ No empty or generic suggestion lists

Recommendation Quality

How Well Does It Perform?

- ▶ About **70% of Top-5 recommendations** are relevant
- ▶ Focuses on **quality over quantity**
- ▶ Recommends movies users are likely to enjoy

What This Means

- ▶ Users trust recommendations
- ▶ Increased likelihood of watching suggested content

Example User Experience

What a User Sees

- ▶ Each recommendation includes:
- ▶ Movie title
- ▶ Genre
- ▶ Expected enjoyment level

Result

- ▶ Clear, personalized, easy-to-choose suggestions
- ▶ Better viewing experience

Business Impact

Why This Project Matters

- ▶ Increased watch time
- ▶ Improved user satisfaction
- ▶ Reduced churn
- ▶ Better content discovery
- ▶ Scalable personalization solution

Strategic Advantage

- ▶ Enhances user loyalty without manual curation

Limitations & Future Improvements

Current Limitations

- ▶ Limited information for brand-new genres
- ▶ No use of actors, directors, or reviews

Future Opportunities

- ▶ Add richer content details
- ▶ Use real-time user behavior
- ▶ Improve personalization depth

Conclusion

Final Takeaways

- ▶ The project addresses a real business problem
- ▶ Personalized recommendations improve engagement and retention
- ▶ The hybrid approach ensures reliability in real-world conditions

Final Outcome

- ▶ A practical, business-ready recommendation system that adds measurable value.

Next step

- ▶ Pilot the recommendation system with a small group of users
- ▶ Collect feedback to measure user satisfaction and engagement
- ▶ Integrate additional data such as genres and viewing behavior
- ▶ Monitor performance and update recommendations regularly

Business Value:

- ▶ These steps ensure the system delivers measurable impact before full deployment.

Thank You

💬 Questions?

📌 Contact Information Names

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